



1. Description

1.1. Project

Project Name	OBD12
Board Name	custom
Generated with:	STM32CubeMX 6.15.0
Date	08/19/2025

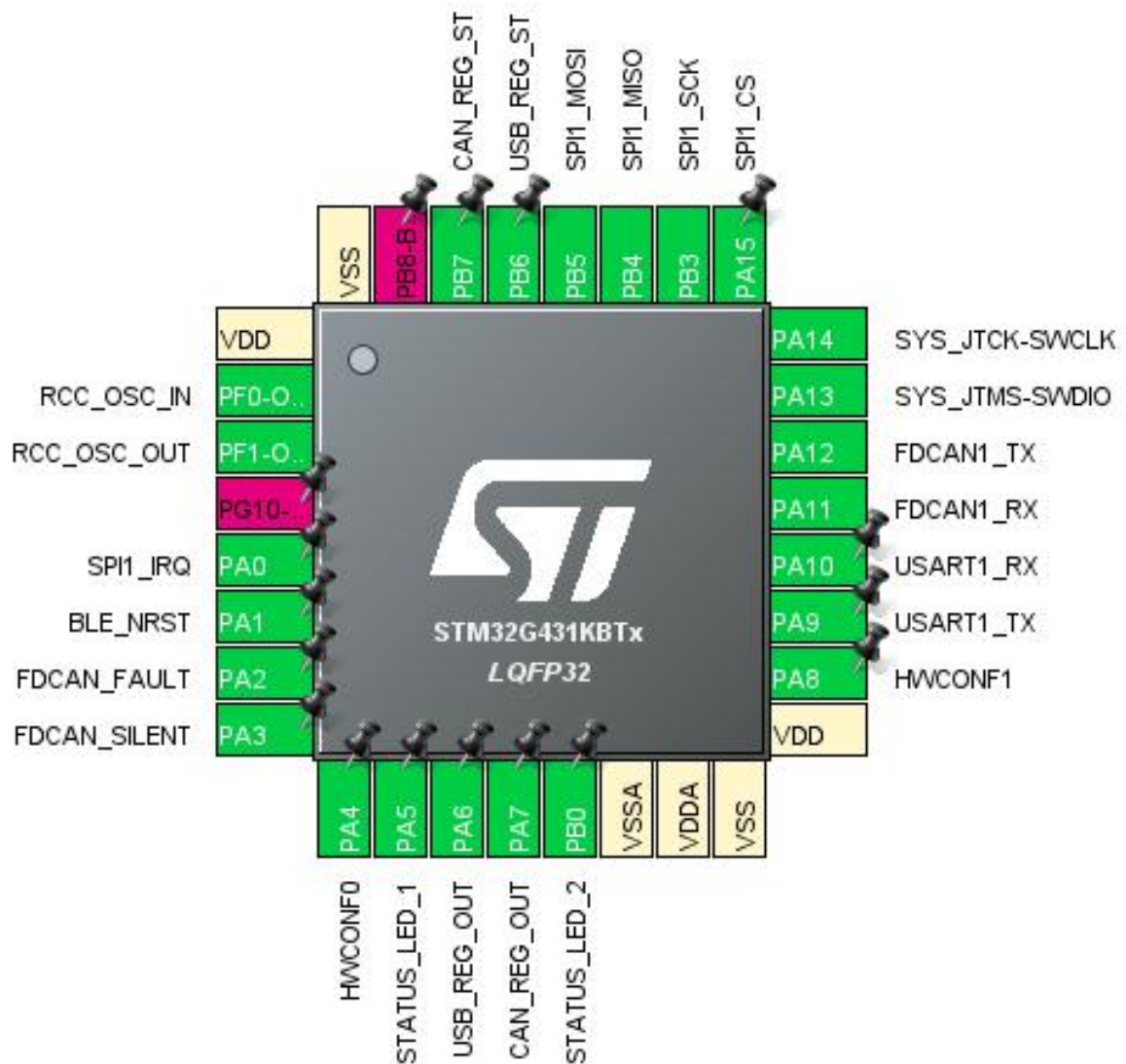
1.2. MCU

MCU Series	STM32G4
MCU Line	STM32G4x1
MCU name	STM32G431KBTx
MCU Package	LQFP32
MCU Pin number	32

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration

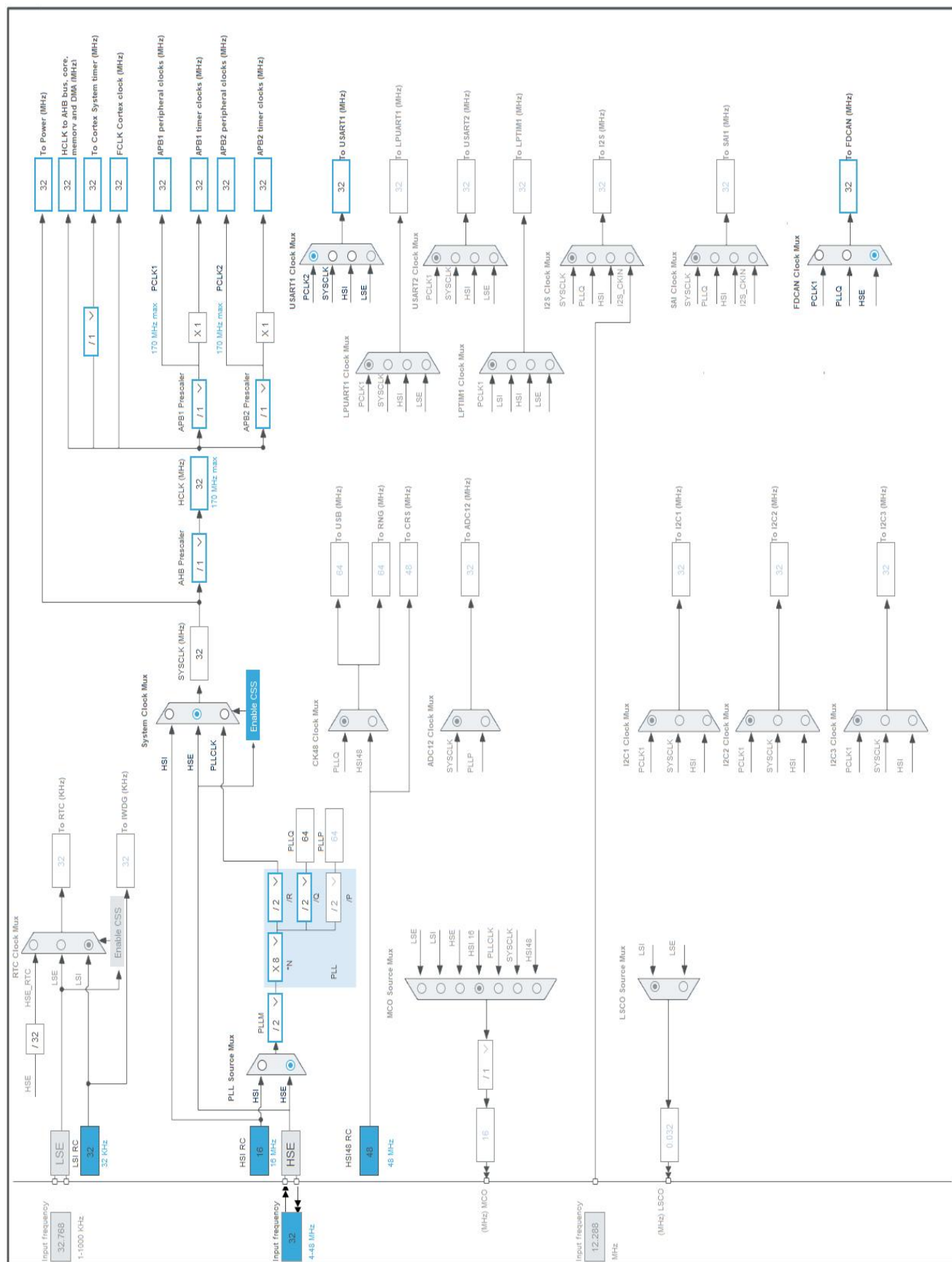


3. Pins Configuration

Pin Number LQFP32	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VDD	Power		
2	PF0-OSC_IN	I/O	RCC_OSC_IN	
3	PF1-OSC_OUT	I/O	RCC_OSC_OUT	
4	PG10-NRST	I/O		
5	PA0	I/O	GPIO_EXTI0	SPI1_IRQ
6	PA1 *	I/O	GPIO_Output	BLE_NRST
7	PA2	I/O	GPIO_EXTI2	FDCAN_FAULT
8	PA3 *	I/O	GPIO_Output	FDCAN_SILENT
9	PA4 *	I/O	GPIO_Input	HWCONF0
10	PA5 *	I/O	GPIO_Output	STATUS_LED_1
11	PA6	I/O	ADC2_IN3	USB_REG_OUT
12	PA7	I/O	ADC2_IN4	CAN_REG_OUT
13	PB0 *	I/O	GPIO_Output	STATUS_LED_2
14	VSSA	Power		
15	VDDA	Power		
16	VSS	Power		
17	VDD	Power		
18	PA8 *	I/O	GPIO_Input	HWCONF1
19	PA9	I/O	USART1_TX	
20	PA10	I/O	USART1_RX	
21	PA11	I/O	FDCAN1_RX	
22	PA12	I/O	FDCAN1_TX	
23	PA13	I/O	SYS_JTMS-SWDIO	
24	PA14	I/O	SYS_JTCK-SWCLK	
25	PA15 *	I/O	GPIO_Output	SPI1_CS
26	PB3	I/O	SPI1_SCK	
27	PB4	I/O	SPI1_MISO	
28	PB5	I/O	SPI1_MOSI	
29	PB6 *	I/O	GPIO_Input	USB_REG_ST
30	PB7 *	I/O	GPIO_Input	CAN_REG_ST
31	PB8-BOOT0	I/O		
32	VSS	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G4
Line	STM32G4x1
MCU	STM32G431KBTx
Datasheet	DS12589 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

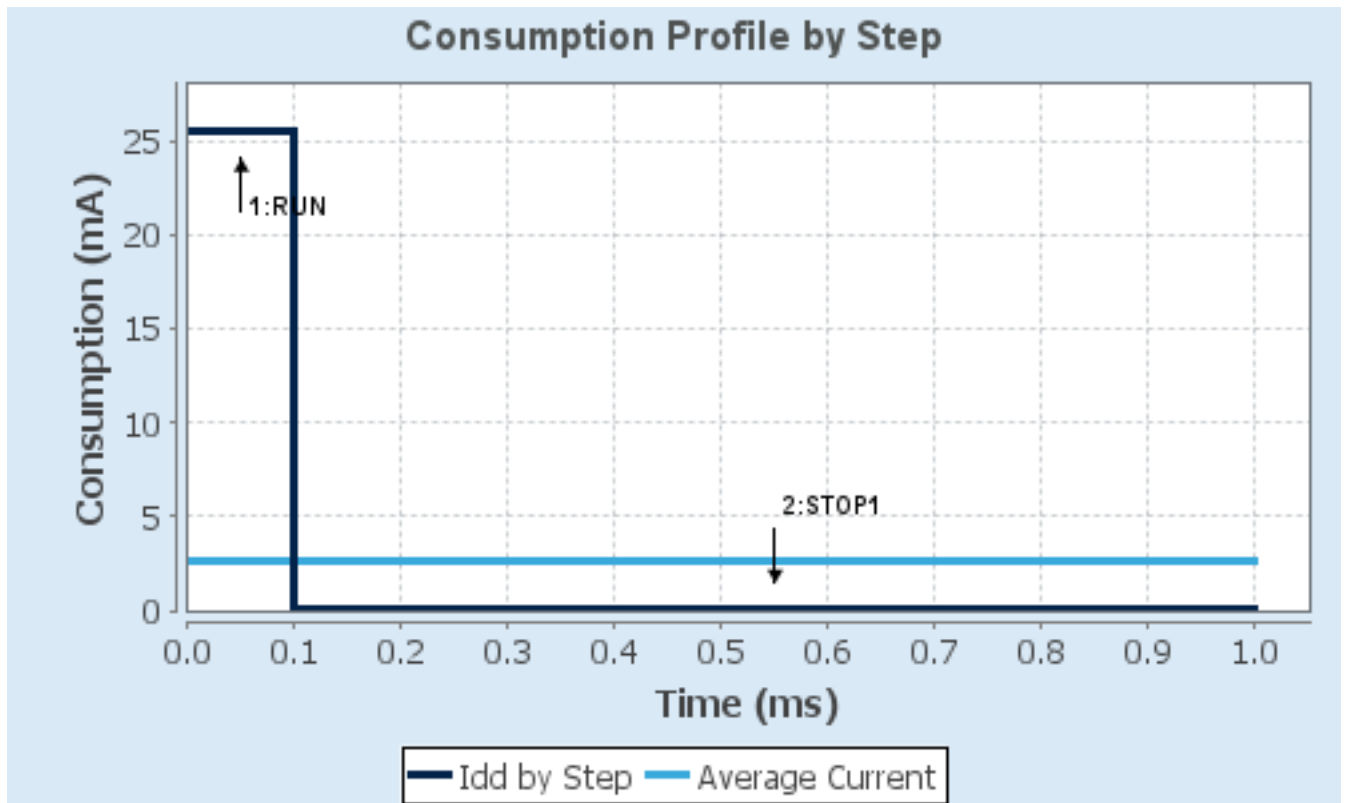
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-Boost	NoRange
Fetch Type	FLASH/ART	NA
CPU Frequency	170 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL CLOCKS OFF
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	25.5 mA	59 μ A
Duration	0.1 ms	0.9 ms
DMIPS	213.0	0.0
Ta Max	124.19	129.99
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	2.6 mA
Battery Life	1 month, 23 days, 22 hours	Average DMIPS	212.5 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	OBD12
Project Folder	C:\Users\samna\Desktop\OBD12\firmware\OBD12
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_G4 V1.6.1
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_ADC2_Init	ADC2
4	MX_USART1_UART_Init	USART1
5	MX_FDCAN1_Init	FDCAN1
6	MX_TIM6_Init	TIM6
7	MX_TIM7_Init	TIM7

3. Peripherals and Middlewares Configuration

3.1. ADC2

IN3: IN3 Single-ended

mode: IN4

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler Synchronous clock mode divided by 2

Resolution ADC 12-bit resolution

Data Alignment Right alignment

Gain Compensation 0

Scan Conversion Mode Disabled

End Of Conversion Selection End of single conversion

Low Power Auto Wait Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

Overrun behaviour Overrun data preserved

ADC_Regular_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 3

Sampling Time 2.5 Cycles

Offset Number No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

3.2. FDCAN1

mode: Activated

3.2.1. Parameter Settings:

Basic Parameters:

Clock Divider	Divide kernel clock by 1
Frame Format	Classic mode
Mode	Normal mode
Auto Retransmission	Disable
Transmit Pause	Disable
Protocol Exception	Disable
Nominal Sync Jump Width	4 *
Data Prescaler	1
Data Sync Jump Width	1
Data Time Seg1	1
Data Time Seg2	1
Std Filters Nbr	1 *
Ext Filters Nbr	0
Tx Fifo Queue Mode	FIFO mode

Bit Timings Parameters:

Nominal Prescaler	1 *
Nominal Time Quantum	31.25 *
Nominal Time Seg1	50 *
Nominal Time Seg2	13 *
Nominal Time for one Bit	2000 *
Nominal Baud Rate	500000 *

3.3. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.3.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Disabled

Data Cache	Enabled
Flash Latency(WS)	1 WS (2 CPU cycle)
RCC Parameters:	
HSI Calibration Value	64
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
Power Parameters:	
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
Peripherals Clock Configuration:	
Generate the peripherals clock configuration	TRUE

3.4. SPI1

Mode: Full-Duplex Master

3.4.1. Parameter Settings:

Basic Parameters:	
Frame Format	Motorola
Data Size	8 Bits *
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	16 *
Baud Rate	2.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	2 Edge *
Advanced Parameters:	
CRC Calculation	Disabled
NSS Signal Type	Software

3.5. SYS

Debug: Serial Wire

Timebase Source: SysTick

mode: save power of non-active UCPD - deactive Dead Battery pull-up

3.6. TIM6

mode: Activated

3.6.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	32000 - 1 *
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	1000 - 1 *
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Trigger Event Selection	Reset (UG bit from TIMx_EGR)
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3.7. TIM7

mode: Activated

3.7.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	32000 - 1 *
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	100 - 1 *
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Trigger Event Selection	Reset (UG bit from TIMx_EGR)
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3.8. USART1

Mode: Asynchronous

3.8.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.9. STMicroelectronics.X-CUBE-BLE2.3.3.0

mode: WirelessJjBlueNRGAa2

3.9.1. Parameter Settings:

Log & Debug:

BLE2_DEBUG

1: Only User messages are shown *

PRINT_CSV_FORMAT

0: .csv format message print disabled

BLUENRG2_DEBUG

0: No debug message

Basic Parameters:

HCI_READ_PACKET_SIZE	128
HCI_MAX_PAYLOAD_SIZE	128
HCI_READ_PACKET_NUM_MAX	10

Connection Parameters (for expert users):

Scan Interval (SCAN_P)	16384
Scan Window (SCAN_L)	16384
Supervision Timeout (SUPERV_TIMEOUT)	60
Min Connection Period (CONN_P1)	40
Max Connection Period (CONN_P2)	40
Min Connection Length (CONN_L1)	2000
Max Connection Length (CONN_L2)	2000
Advertising Type (ADV_DATA_TYPE)	Connectable Undirected Advertising (ADV_IND)
Min Advertising Interval (ADV_INTERV_MIN)	400 *
Max Advertising Interval (ADV_INTERV_MAX)	800 *

Min Connection Event Interval (L2CAP_INTERV_MIN)	9
Max Connection Event Interval (L2CAP_INTERV_MAX)	20
Timeout Multiplier (L2CAP_TIMEOUT_MULTIPLIER)	600

3.9.2. Platform Settings:

Exti Line	PA0
BUS IO driver	SPI1
CS Line	PA15
Reset Line	PA1

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC2	PA6	ADC2_IN3	Analog mode	Pull-down *	n/a	USB_REG_OUT
	PA7	ADC2_IN4	Analog mode	Pull-down *	n/a	CAN_REG_OUT
FDCAN1	PA11	FDCAN1_RX	Alternate Function Push Pull	No pull-up and no pull-down	High *	
	PA12	FDCAN1_TX	Alternate Function Push Pull	No pull-up and no pull-down	High *	
RCC	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SPI1	PB3	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	
	PB4	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	
	PB5	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	
USART1	PA9	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	
	PA10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	
GPIO	PA0	GPIO_EXTI0	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	SPI1_IRQ
	PA1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BLE_NRST
	PA2	GPIO_EXTI2	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	FDCAN_FAULT
	PA3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FDCAN_SILENT
	PA4	GPIO_Input	Input mode	Pull-down *	n/a	HWCONF0
	PA5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	STATUS_LED_1
	PB0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	STATUS_LED_2
	PA8	GPIO_Input	Input mode	Pull-down *	n/a	HWCONF1
	PA15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI1_CS
	PB6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USB_REG_ST
	PB7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	CAN_REG_ST

4.2. DMA configuration

nothing configured in DMA service

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	0	0
EXTI line0 interrupt	true	1	0
EXTI line2 interrupt	true	1	0
FDCAN1 interrupt 0	true	1	0
TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts	true	2	0
TIM7 global interrupt	true	0	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 and ADC2 global interrupt	unused		
FDCAN1 interrupt 1	unused		
SPI1 global interrupt	unused		
USART1 global interrupt / USART1 wake-up interrupt through EXTI line 25	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
EXTI line0 interrupt	false	true	true
EXTI line2 interrupt	false	true	true
FDCAN1 interrupt 0	false	true	true
TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts	false	true	true
TIM7 global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware

Software Packs

X-CUBE-BLE2 

System Core

Analog

Timers

Connectivity

Multimedia

Security

Computing

Utilities

DMA

ADC2 

TIM6 

FDCAI1 

GPIO 

TIM7 

SP1 

IIVIC 

USART1 

RCC 

SYS 

6. Software Pack Report

6.1. Software Pack selected

Vendor	Name	Version	Component
STMicroelectronics	X-CUBE-BLE2	3.3.0	Class : Wireless Group : BlueNRG-2 SubGroup : Controller Version : 3.3.0 Class : Wireless Group : BlueNRG-2 SubGroup : HCI_TL Variant : Basic Version : 3.3.0 Class : Wireless Group : BlueNRG-2 SubGroup : HCI_TL_INTERFACE Variant : UserBoard Version : 3.3.0 Class : Wireless Group : BlueNRG-2 SubGroup : Utils Version : 3.3.0

7. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g4_svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32g4.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/fldpstpf11120.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

- Application Notes https://www.st.com/resource/en/application_note/an2606-stm32-microcontroller-system-memory-boot-mode-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4232-getting-started-with-analog-comparators-for-stm32f3-series-and-stm32g4-series-devices-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4296-use-stm32f3stm32g4-ccm-sram-with-iar-embedded-workbench-keil-mdkarm-stmicroelectronics-stm32cubeide-and-other-gnubased-toolchains-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-

digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5093-getting-started-with-stm32g4-series--hardware-development-boards-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5306-operational-amplifier-opamp-usage-in-stm32g4-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5310-guideline-for-using-analog-features-of-stm32g4-series-versus-stm32f3-series-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5315-stm32cube-firmware-examples-for-stm32g4-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5346-stm32g4-adc-use-tips-and-recommendations-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5094-migrating-between-stm32f334303-lines-and-stm32g431xxg474xxg491xx-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5738-stm32g4-series-lifetime-estimates-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5325-how-to-use-the-cordic-to-perform-mathematical-functions-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to

usb-typec-power-delivery-for-stm32-mcus-and-mpus-
stmicroelectronics.pdf

Application Notes [https://www.st.com/resource/en/application_note/an4894-how-to-use-
eeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4894-how-to-use-
eeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-
oversampling-techniques-to-improve-signal-to-noise-ratio-on-stm32-mcus-
stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-
oversampling-techniques-to-improve-signal-to-noise-ratio-on-stm32-mcus-
stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application_note/an5036-guidelines-for-
thermal-management-on-stm32-applications-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5036-guidelines-for-
thermal-management-on-stm32-applications-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application_note/an5405-how-to-use-
fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5405-how-to-use-
fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application_note/an5978-introduction-to-
mb1971-llc-hat-12-v-to-75-v1-a-for-f334-g474-nucleo-board-
stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5978-introduction-to-
mb1971-llc-hat-12-v-to-75-v1-a-for-f334-g474-nucleo-board-
stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application_note/an5690-how-to-use-
vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an5690-how-to-use-
vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf)

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